

Sound of Sirens as Acoustic Landmarks within the European Union

Juan Manuel Rodríguez Mejía*, Christian Kehling*, Kashan Sardar*, and Stephan Werner*

**Technische Universität Ilmenau, Ilmenau, Germany*

Electronic Media Technology Group

Email: juan-manuel.rodriguez-mejia@tu-ilmenau.de

Abstract—Siren sounds share properties with musical signals, such as recognizable pitches and spectral similarities.

In applications involving deep learning, data augmentation plays a crucial role in increasing diversity and quantity within data, which enhances model robustness. This paper introduces a VGG-based deep learning model designed to classify emergency siren sounds, distinguishing police, ambulance, and fire truck sirens in France, Germany, Italy, and Spain, achieving significantly accuracy with fewer parameters than in the ResNet model [1]. We faced challenges like dataset imbalance and spectral variability with targeted data augmentation techniques. Our study underscores the importance of siren classification within the European Union, identifying it as a potential landmark for verifying acoustic similarities and differences between emergency signals.

Index Terms—Siren detection, neural networks, deep learning, VGG, emergency vehicle classification, urban safety, data augmentation.

I. INTRODUCTION

In busy urban settings, the piercing sound of sirens rises above the city noise, each with its own pitch and harmonics, signaling the approach of emergency vehicles like police cars, ambulances, and fire trucks. These sounds play a critical role in alerting people to potential danger and prompting quick responses [2]. Recognizing the type and source of these sirens is essential for effective emergency management and public safety.

Recent advances in deep learning have changed the way we process these acoustic signals, making it possible to detect and classify siren sounds [2]. In [1] the concept of ResNet (Residual Network) model was introduced, which focuses on classifying sirens based on acoustic landmarks. We propose using a VGG-based (Visual Geometry Group) architecture to further improve siren identification. A major challenge in siren classification is the variability and imbalance in the data. Siren sounds can vary widely, making it difficult to develop models that work consistently in different urban settings [3]. To address this, robust data augmentation techniques are crucial to ensure that our models are well generalized and do not overfit [4].

Our study aims to classify sirens to gain a better understanding of sound event scenes, which could benefit various applications, including urban noise monitoring, content verification, and enhancing how autonomous vehicles respond to emerging signals [?]. By improving the accuracy of siren detection and reducing the number of parameters used in our model compared to this paper, our approach seeks to enhance the ability to differentiate between country-specific siren classes within the European Union (France, Germany, Italy, and Spain). To achieve this, we introduce the Siren Classification: Police, Ambulance, and Firefighter model (SICLAPAF).

In this paper, we delve into the complexity of classifying siren sounds using deep learning methods. We examine how humans perceive and recognize these sounds, evaluate the effectiveness of the VGG-like model compared to other approaches, and discuss implications for future research in acoustic signal processing and urban safety systems [5]. Through these efforts, we hope to advance technologies for emergency response and refine our understanding of urban acoustics.

II. RELATED WORK

In recent years, deep learning techniques have significantly advanced acoustic signal processing, particularly in classifying environmental sounds. Commonly used architectures include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and their variants. CNNs excel at capturing spatial hierarchies in data, which makes them

ideal for tasks based on images and spectrograms [2]. On the other hand, RNNs are adept at processing sequential data, leveraging temporal dependencies for tasks such as time series analysis and sequential classification [6].

This current paper contrasts with the existing literature by focusing on the classification of emergency sirens in the European Union using the dataset introduced in [1]. Initially proposed for siren classification based on acoustic landmarks, the ResNet model serves as the foundation for our study, where we enhance this approach by integrating the VGG-like architecture known for its effectiveness in extracting features from spectrograms [3]. A significant contribution of our work addresses dataset imbalances and the inherent variability of urban soundscapes. Data augmentation techniques like pitch and time-frequency shifts or adding background noise can help address data imbalance and improve model robustness against variations in siren types and background noise, as shown in [4]. However, for this work, we chose to implement an organic symmetrical dataset to better suit our specific application and enhance the reliability of urban safety systems and emergency response technologies.

III. METHODOLOGY

A. Dataset and Data Representation

1) *Dataset*: Table I presents the distribution of samples in the training and testing datasets for each country and category. Each dataset comprises samples from four countries: France, Germany, Italy, and Spain. For each country, there are samples for three types of siren: ambulance, firefighter, and police.

TABLE I
TRAINING AND TESTING DATASETS FOR MULTIPLE COUNTRIES

Countries	Classes	Train Samples	Test Samples
France, Germany, Italy, Spain	Total	93	30
	Ambulance	31	10
	Firefighter	31	10
	Police	31	10

For training and testing datasets, the audio sampling rate is set at 44.1 kHz, with each sample having a duration of 5 seconds. A Hann window shape is used, with a window and FFT size of 2048 samples. The hop size is 512 samples, which corresponds to approximately 11.61 ms at a 44.1 kHz sample rate.

2) *Data Representation*: A contribution of our work addresses dataset imbalances and the inherent variability of urban soundscapes. By augmenting the dataset with techniques that involve mix-up, pitch shifts or adding background noise, enabling to generate synthetic data.

Key parameters such as hop-size, bits per second (bps), and the Per-Channel Energy Normalization (PCEN) method significantly impact model performance. PCEN and Data Representation Techniques configurations, depending on the model, are shown in the table II.

TABLE II
COMPARISON OF RESNET AND VGG MODELS

Model	PCEN Usage	Data Representation Techniques
ResNet	Yes	Time-frequency transformations
SICLAPAF-VGG	No	Mel-Spectrogram augmentation

3) *Per-Channel Energy Normalization*: This technique improves feature extraction in noisy environments by adaptively smoothing the energy of each spectrogram channel and compressing its dynamic range, providing robust representations for applications such as speech recognition and sound events in fluctuating noise conditions [7]. PCEN normalization is particularly effective in enhancing the contrast of acoustic features, addressing a better classification of sirens whilst the background noise levels can be attenuated.

B. Deep Learning Architectures

In general terms, any deep learning architecture can be implemented for the classification of emergency vehicle sirens. Those can include CNNs and their variants, such as ResNet and the VGG architecture. These models are based on learning hierarchical features from spectrogram representations of audio signals, which are crucial to distinguish between different types of emergency sirens.

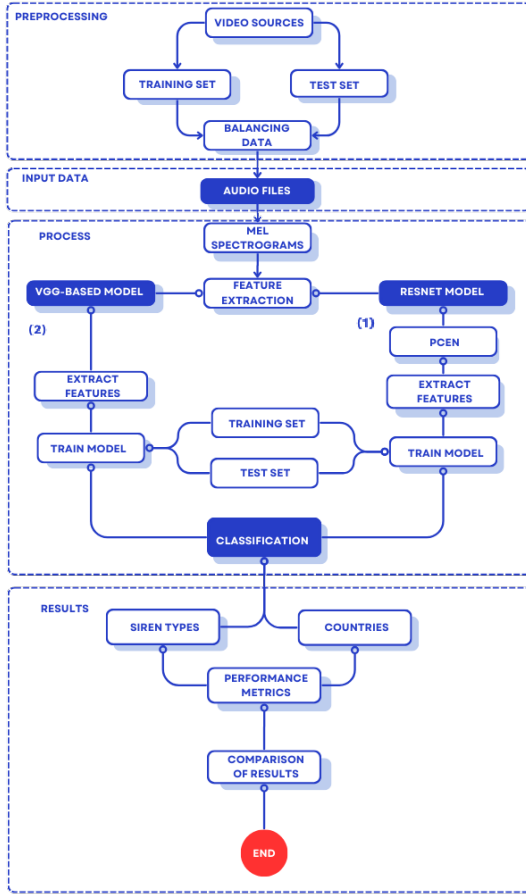


Fig. 1. Block Diagram of the entire processing for siren classification with two models ResNet - (1) path and SICLAPAF - (2) path

The flow chart in Figure 1 corresponds to the pipelines when deep learning is applied to classify emergency sirens per region and siren type using ResNet and VGG architectures. The diagram starts with an audio file that needs to be understood as a spectrogram by the two models used.

The two models are: (1) This model uses per channel energy normalization (PCEN) [7] to reduce environmental noise, and (2) the SICLAPAF framework uses the geometrical features of the spectrograms obtained from the sirens. For comparative analysis, both models generate confusion matrices with accuracy metrics. These matrices allow us to visualize the classification performance across different classes, such as police, firefighter, and ambulance sirens, in four countries: France, Germany, Italy, and Spain.

C. SICLAPAF model

Figure 2 illustrates the architecture of the VGG-like model implemented in this study. The model features a convolutional block consisting of six layers with filters set to 32, 32, 32,

64, 64, and 128, respectively, each utilizing a 3×3 filter size. The pooling operations are set with a size of 2×2 , effectively downsampling the feature maps and enhancing the model's ability to capture spatial hierarchies.

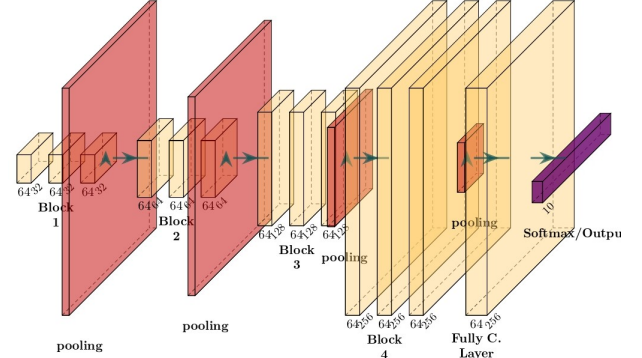


Fig. 2. Overview of VGG Model Architecture

ReLU activation functions, which output values in the range $[0, \infty)$, are applied in the network to introduce non-linearity. The dense layers have 256 units and incorporate a dropout rate of 0.5, aimed at mitigating overfitting during training. These layers also employ ReLU activation to maintain the model's capacity to learn complex patterns. The number of units in the output layer varies according to the number of classes for the classification task, uses Softmax activation to generate class probabilities. The training regimen employs the Adam optimizer with a learning rate of 0.001 and a batch size of 32, optimizing the categorical cross-entropy loss function. The total number of parameters in this model is approximately 220,000, with evaluation metrics focusing on accuracy to assess performance effectively.

D. ResNet Model

The ResNet model used in this study is structured to effectively capture complex features. As depicted in Figure 3, the model begins with a convolutional block comprising 64 filters with a kernel size of 5×5 , activated by the ReLU function. This initial layer serves as a feature extractor, transforming the input data into informative feature maps. Following this, the architecture incorporates four residual blocks, featuring filter sizes of 64, 64, 128, and 128, respectively. These blocks leverage skip connections, allowing the model to learn residual mappings instead of direct mappings, which facilitates the training of deeper networks and alleviates the vanishing gradient problem. The diagram provides a clear visual representation of these components and their interconnections, highlighting how the architecture processes information through its layers. Figure 3 illustrates the architecture of the ResNet model. The model includes global average pooling and comprises around 800k parameters in total.

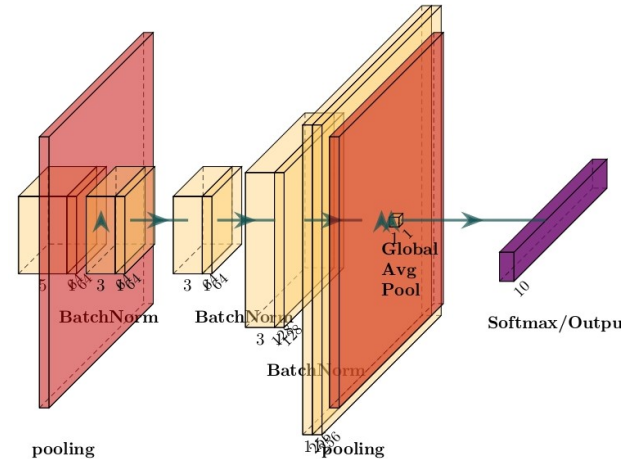


Fig. 3. Overview of ResNet Model Architecture

The architecture is completed with a global average pooling layer that condenses the feature maps into a single vector, significantly reducing dimensionality while preserving essential spatial information. This is followed by a softmax layer that classifies the output into ten distinct categories. With approximately 800,000 parameters, the model is capable of learning a wide range of features effectively. The combination of the diagram and the detailed description of the model's architecture offers a comprehensive understanding of how the ResNet model is designed for efficient signal processing and robust learning, providing valuable insights into its experimental setup.

IV. EVALUATION AND RESULTS

To evaluate model performance, confusion matrices and accuracy metrics are computed based on predictions made on a test dataset. These metrics provide insights into the model's ability to correctly classify different siren types and their regional variations.

A. Confusion Matrices

For the ResNet and VGG models, the confusion matrices show the country classes in the first plot, and given is a percentual value that estimates the identification of the labels.

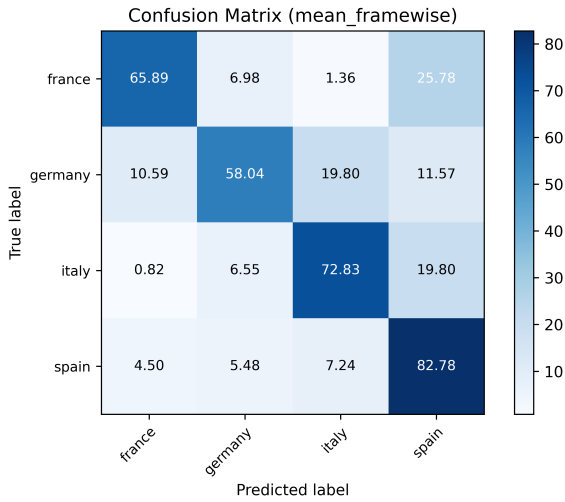


Fig. 4. Confusion Matrix for ResNet Model (ResNet) Across All Countries

Figure 4 illustrates the performance of the ResNet model in classifying emergency vehicle sirens into different categories using a confusion matrix. The matrix displays the true label classes versus the predicted classes of the four countries, allowing assessment of the model's ability to distinguish between countries. The model shows higher accuracies in Italy and Spain, while the accuracies for France and Germany are lower with values of 0.65 and 0.58, respectively.

Figure 5 represents the performance of the VGG model (SICLAPAF) in classifying four countries as categories. The results are similar to the previous model; however, there is a wrong identification of siren classes from France as Italy. And the best values are for Germany and Spain with 0.6 accuracy.

V. CONCLUSION

In conclusion, maintaining balanced data across classes and subclasses is essential for robust model training, and selecting optimal parameters, such as learning rate, batch size, and epoch count, is crucial for effective performance in NN and CRNN models. Although the confusion matrix provides valuable information on classification outcomes, the F1 score serves as a more comprehensive metric that integrates accuracy. This metric is particularly relevant for assessing the effectiveness of the model in binary and multiclass scenarios, where class imbalance may be present. However, F1 scores have not yet been fully evaluated, and these calculations are planned for future analysis.

With respect to the classification of siren types, the results are less accurate than those of countries. For both models is more accurate to identify firefighter siren. This may arise

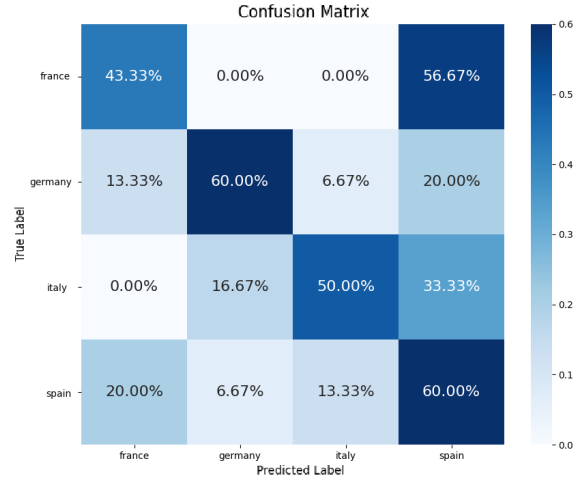


Fig. 5. Confusion Matrix for VGG Model (SICLAPAF) Across All Countries

from high similarity, variance, or minimal differences in sound characteristics between the three siren-type classes. Additionally, the limited size of the siren dataset contributes to these challenges, as models may struggle to generalize with fewer examples per class.

Despite these challenges, simpler models, such as the VGG-based architecture, have shown promising accuracy (exceeding 50%) even with a relatively low parameter count. This result highlights the potential of streamlined architectures in effective classification tasks. The reduced parameter count in the VGG-based model is an advantage because its high performance can still be achieved without the computational demands of more complex models.

VI. FUTURE WORK

Future work will focus on incrementing the dataset gradually, ensuring balanced representation across classes without relying on data augmentation, thus allowing for equal amounts of data in subsets such as siren type and country. In addition, models such as Kolmogorov–Arnold Networks and Reservoir Computing will be explored for their potential in siren classification. These models offer distinct advantages in capturing complex temporal dependencies and dynamic patterns, which are crucial for accurately distinguishing between siren types and understanding some variations, potentially leading to significant improvements in classification accuracy, using fewer parameters.

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